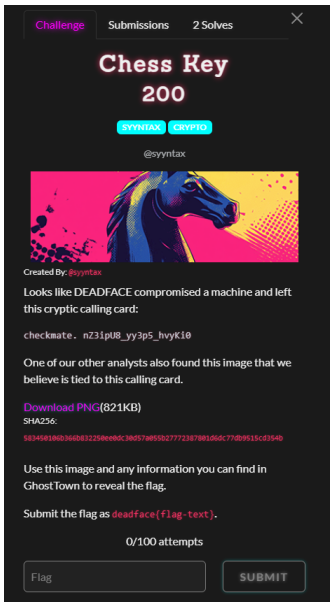
Solving “Chess Key” (200 pts - Crypto/OSINT)

A hybrid solve combining a custom chessboard substitution table, public chess-game research, and Vigenere-style decryption.



Challenge page used during the solve.

1. The Challenge

The ciphertext was paired with a letter-filled chessboard image and a GhostTown clue referencing Magnus Carlsen. The intended key had to be reconstructed from a real 2026 game.

2. Read the Board as Coordinates

Treat the image as a normal chessboard: a8 is the upper-left square and h1 the lower-right. Each destination square maps to the letter written in that square.

```
Examples from the supplied board:
c4 -> L
g3 -> S
g2 -> G
f3 -> F
h4 -> I
```

3. Follow the OSINT Clue

The GhostTown clue narrows the search to a 2026 Magnus Carlsen game using the English Opening. Using Magnus’s move destination squares in order produces the letter key needed for the cipher.

4. Decrypt the Calling Card

Apply the chess-derived key to the alphabetic characters in the ciphertext while leaving digits, spaces, and underscores unchanged.

```
Ciphertext: nZ3ipU8 yy3p5_hvyKi0
Plaintext: cH3ckM8 ch3s5_cryPt0
```

5. The Flag

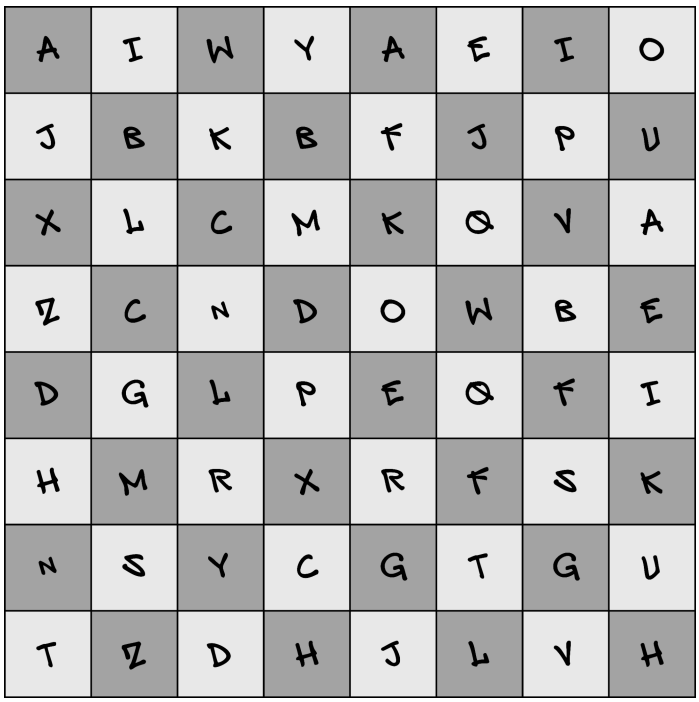
```
deadface{cH3ckM8_ch3s5_cryPt0}
```

6. General Method

• Separate the OSINT step (identify the game) from the crypto step (derive and apply the key).

• Use destination-square notation consistently when reading the custom board.

• Preserve nonletters when decrypting mixed leetspeak ciphertext.



The custom chessboard substitution table supplied with the challenge.

Summary

Field	Result
Category	Crypto + OSINT
External clue	2026 Magnus Carlsen English Opening game
Key source	Move destination squares -> board letters
Flag	deadface{cH3ckM8_ch3s5_cryPt0}